Let's now look at the characteristics and adoption patterns of a PaaS platform.

PaaS adoption patterns

A cloud computing service must satisfy five essential characteristics – it should be available as an on-demand service, have access to a network, have resource pooling, have elasticity and offer measured services. To achieve these essential characteristics, cloud computing provides three kinds of service models: Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service. Figure 1 shows the cloud computing PaaS adoption pattern.

The key business drivers of PaaS adoption are:

- Reduced capex and opex to deliver business services
- Minimised IT costs by improving delivery time and the quality of the app
- Increased flexibility and integration between middleware components

Simple PaaS is the entry point into the PaaS space. It allows the provisioning of application services and exposing them as services into a self-service catalogue, automating the deployment and metering the resources used by this service.

Manage PaaS manages the SLA and QoS aspects of the provisioned applications like resiliency, application performance, security, etc.

Programming PaaS allows the applications to integrate with external applications or public clouds in order to implement auto-scaling and cloud bursting scenarios.

Process oriented PaaS allows the implementation of a DevOps process by creating a continuous delivery flow that automates the building, testing and delivery of an application into a cloud environment.

In addition to the above adoption patterns, the other variations of PaaS are listed below. These variations might align to one of the patterns explained above.

iPaaS: Integration Platform as a Service (iPaaS) is a suite of cloud services enabling the development, execution and governance of integration flows connecting any combination of on-premises and cloud-based processes, services, applications and data within individual or across multiple organisations. Examples include MuleSoft CloudHub, BizTalk Services, etc.

mPaaS: Mobile Platform as a Service (mPaaS) is the provision of an interactive development environment (IDE) to create mobile apps. mPaaS supports multiple mobile operating platforms.

dbPaaS: Database Platform as a Service is an on-demand, secure and scalable self-service database platform that automates the provisioning and administration of databases. When using dbPaaS, database assets make scaling easier and offer better reliability.

IoTPaaS: This provides common infrastructure to enable communication, security, analytics and the management for heterogeneous IoT topologies. It provides simpler and more agile models for building IoT solutions.

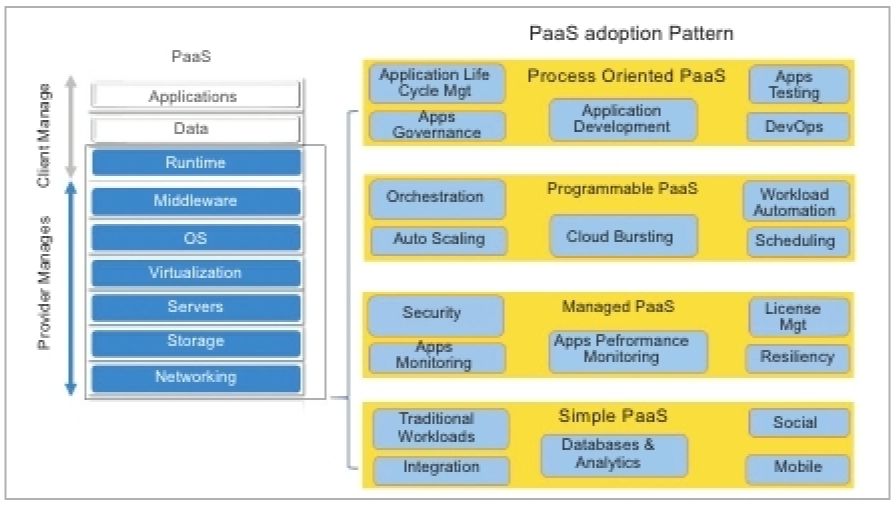

Figure 1: PaaS adoption patterns

bpmPaaS: This is a complete pre-integrated BPM platform hosted in the cloud and delivered as a service. It is leveraged for the development and execution of business process and workflow centric applications across enterprises. Examples are Pega Cloud, OpenText Cordys Cloud, etc.

Characteristics of PaaS

The following are some basic characteristics of PaaS:

- Offers services to develop, test, deploy, host and maintain applications in the same integrated development environment
- Multi-tenant architecture in which multiple concurrent users use the same development application
- Built-in scalability of deployed software, including load balancing and failover
- Integration with heterogeneous platforms and systems
- Support for collaboration within the development team
- Tools to handle billing and subscription management

PaaS reference architecture

The PaaS cloud delivery model exclusively focuses on application development and delivery. PaaS advocates a new kind of development based on native support for concepts like agile development, unit testing, continuous integration and automatic scaling, while providing a range of middleware capabilities. Applications developed on these can be deployed as services and managed across thousands of application instances, which are referred to as nodes.

The significant PaaS architecture components are described below.

Application layer: This comprises reusable components for the reconstruction of native components in the cloud.

Resilience: The circuit breaker pattern is commonly used to ensure that when there is a failure, a failed service does not adversely affect the entire system. A circuit breaker is an always-live system keeping watch over dependency invocations. In case of a high failure rate, the circuit breaker stops the calls from going through for a short period of time.

Service discovery: In a modern, cloud-based microservices application, service instances have dynamically assigned network locations. Moreover, service instances